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PLM-YOLOv5: Improved YOLOv5 for Pig Detection in Livestock Monitoring

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***Abstract*—With the rapid development of smart agriculture, the demand for pig monitoring has been increasing, particularly in disease prevention and behavioral analysis. Real-time monitoring of pig behavior is essential for improving farming efficiency and ensuring animal welfare. However, challenges such as pigs' varying body sizes, movement, and complex environments make accurate detection and tracking difficult. To address these issues, this paper proposes an improved YOLOv5 model, named PLM-YOLOv5, specifically designed for pig detection in livestock monitoring. First, we add an additional detection head to enhance the model's capability to detect smaller pigs effectively. Next, we integrate the Convolutional Block Attention Module (CBAM) to enhance feature extraction, particularly in handling complex backgrounds. Finally, we utilize a self-collected dataset of over 1,400 images for model training and validation, ensuring the model adapts to real-world livestock environments. Experimental results show that PLM-YOLOv5 achieves significant improvements in pig detection tasks, with the mean Average Precision (mAP) increasing by approximately 9% compared to the original YOLOv5. Additionally, training and validation loss curves demonstrate the model's stability and convergence. These results confirm the effectiveness and reliability of PLM-YOLOv5 for real-world applications, offering a robust solution for smart agriculture.**

Keywords-YOLOv5, Livestock Monitoring, Smart Agriculture, CBAM, Pig Detection

I. INTRODUCTION

In the modern agricultural and farming industry, precise management and monitoring are crucial for improving productivity and ensuring animal health and welfare. As one of the most important livestock species, pigs are crucial for maintaining the stability and sustainability of the meat supply chain. With technological advancements, automated monitoring systems have become essential for analyzing pig behavior, preventing diseases, and enhancing farming efficiency. Pig detection technology is a core component of such systems, directly contributing to achieving these objectives.

Pig detection technology has several applications in livestock management. First, behavior monitoring enables the early detection of anomalies, such as reduced appetite or diminished activity, which could indicate early signs of illness [1, 2]. Second, tracking the growth of pigs assists farmers in optimizing feed formulations and farming strategies, improving growth rates and meat quality [3, 4]. Finally, reproductive management benefits from pig detection systems, as monitoring the estrous cycles of sows enhances reproductive efficiency and litter sizes [5, 6].

Despite progress in computer vision and machine learning, challenges remain in applying these technologies to pig detection. Small-object detection, such as identifying piglets in crowded and complex environments, is particularly difficult. Additionally, environmental factors like varying lighting conditions, occlusions, and cluttered backgrounds pose significant challenges to detection accuracy. Moreover, the need for real-time data analysis in large-scale farming operations creates high demands for computational efficiency [7, 8].

To address these challenges, this work proposes an improved YOLOv5 [9] model, PLM-YOLOv5, specifically designed for pig detection in livestock monitoring systems. The model incorporates several innovations. First, an additional detection head was introduced specifically for small-object detection, improving the model's accuracy in detecting piglets. Second, the Convolutional Block Attention Module (CBAM) was integrated to strengthen feature extraction, particularly in complex backgrounds. Third, a self-collected dataset of over 1,400 images, representing diverse farming conditions, was used to train and validate the model, improving its generalization and robustness.

Experimental results demonstrate that PLM-YOLOv5 outperforms the original YOLOv5 [9], achieving approximately 9% higher mean Average Precision (mAP) in pig detection tasks. These findings validate the effectiveness of the proposed approach, providing strong technical support for intelligent livestock monitoring systems and contributing to the advancement of smart agriculture.

The primary contributions of this work are outlined as follows:

- Additional detection head: We introduced a new detection head to enhance the accuracy of small-object detection, especially for piglets in challenging environments.
- CBAM integration: The inclusion of the Convolutional Block Attention Module enhances the model's ability to emphasize key features in challenging backgrounds.
- Custom dataset: A self-collected dataset of over 1,400 diverse images was used to ensure robust model performance across various scenarios.

II. RELATED WORKS

A. Data Augmentation

Data augmentation is an essential technique in deep learning to enhance the robustness and generalization of models, particularly when dealing with limited datasets. Augmentation increases the diversity of training samples by applying transformations such as scaling, rotation, flipping, cropping, and color adjustments. These methods have been widely used to address issues like overfitting and poor model performance on unseen data. Advanced methods such as Mosaic augmentation and CutMix [10] have further expanded the possibilities by combining or blending multiple images to enrich feature diversity [11, 12]. For instance, Mosaic merges four distinct images into one composite, allowing models to learn contextual relationships among objects in varying locations. CutMix, on the other hand, blends regions from different samples, forcing models to focus on multiple areas of interest within a single image [13].

Pig detection tasks often face challenges due to the variability in farm environments, such as lighting changes, shadows, and background clutter. Standard augmentation techniques are often insufficient to address these issues. To overcome this, researchers have developed domain-specific augmentations, such as simulating occlusions caused by fencing or varying the intensity of artificial light [14]. Additionally, augmentation methods can help deal with class imbalance, especially when certain pig sizes or postures dominate the dataset. Custom augmentations tailored for agricultural detection tasks have been shown to significantly improve accuracy in challenging environments, such as pig monitoring [15].

Our work leverages data augmentation to enrich the diversity of our self-collected dataset. This includes applying advanced techniques like Mosaic and domain-specific transformations to better represent real-world pig farming conditions. These efforts ensure that the model can generalize effectively to diverse environments, making it more reliable in practical applications. By incorporating these strategies, we tackle the unique challenges of small-object detection and cluttered backgrounds, contributing to a more robust pig monitoring system.

B. Object Detection

Pig detection plays a vital role in modern smart farming by enabling automated monitoring of animal behavior, growth, and welfare. Traditional methods relied heavily on manual observation and static camera setups, which were time-consuming, prone to errors, and lacked scalability. With the advent of deep learning, object detection algorithms such as Faster R-CNN [16] and YOLO [17] have transformed the field by offering faster and more accurate solutions [18, 19]. These methods detect and classify objects in images, making them highly suitable for real-time pig monitoring in large-scale farms.

YOLO-based models [17], in particular, have emerged as a preferred choice due to their balance between speed and precision. YOLOv3 [20] and YOLOv4 [21] have been successfully applied in livestock monitoring scenarios, where their ability to process images in real-time meets the high demands of large-scale farms [22]. However, detecting pigs presents unique challenges. Small piglets, for example, are often hard to distinguish from background elements, especially in cluttered or low-light environments. Chen et al. addressed this issue by integrating a density-aware module into YOLOv3 [20], enhancing its ability to detect pigs in crowded pens. Similarly, Zhang et al. proposed a lightweight YOLO model tailored for piglet detection, which optimized computational efficiency without sacrificing accuracy [23].

Recent advances have also incorporated attention mechanisms to improve feature extraction in challenging environments. Modules like SE (Squeeze-and-Excitation) and CBAM (Convolutional Block Attention Module) have been added to detection frameworks to focus on the most relevant features of pigs, such as shape and texture [24].

Building on these advancements, our work introduces an extra detection head specifically for small-object detection and integrates CBAM to enhance feature representation. As shown in Figure 1, the improved YOLOv5 [9] architecture consists of a Backbone (CSPDarknet53 [25]) for feature extraction, a Neck (PANet [26]) for feature fusion, and a CBAM module to refine the features before passing them to the detection heads. The model includes detection heads designed to detect pigs of various sizes, with an additional detection head specifically for small pigs, addressing the challenges of detecting small piglets in complex environments. This approach helps fill the gaps in existing studies, such as limited datasets and suboptimal performance in small-object scenarios, advancing the state of pig detection in real-world farming environments.

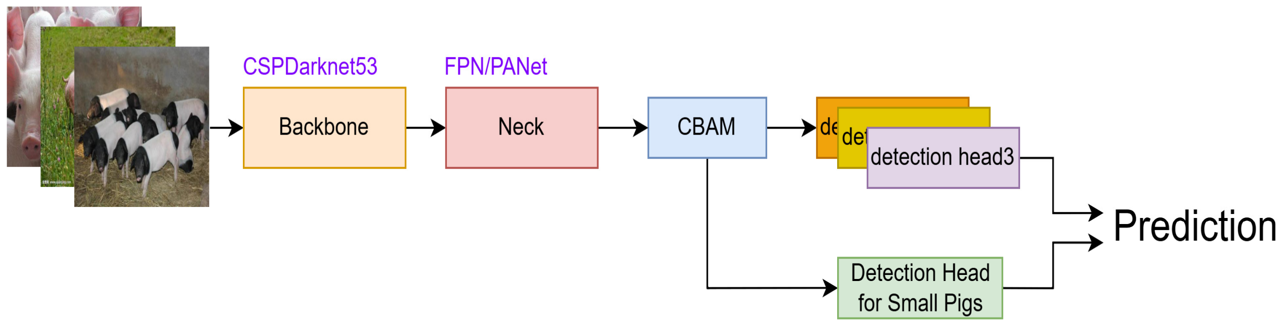

Figure 1. Improved YOLOv5 Architecture for Pig Detection. This diagram illustrates the architecture of the improved YOLOv5 model for pig detection. The system consists of four key components: the Backbone (CSPDarknet53), the Neck (PANet), the CBAM block for enhanced feature extraction, and multiple detection heads (detection head1, detection head2, detection head3), which are used to detect pigs of different sizes, with an additional head for small pigs. Arrows show the flow of features through the model, with each head detecting pigs at various scales.

III. METHODOLOGY

A. *Overview of YOLOv5*

YOLOv5 [9] is a state-of-the-art, real-time object detection model known for its ability to balance speed and accuracy. It is based on a fully convolutional network architecture, making it highly efficient in detecting objects from images. The model consists of three primary components: the Backbone, the Neck, and the Head.

The Backbone (typically CSPDarknet53 [25] in YOLOv5 [9]) is responsible for feature extraction, where it processes the input image and extracts useful features for further analysis. The Neck (comprising PANet [26]) is tasked with fusing features from different scales, allowing the model to effectively detect objects of various sizes. Finally, the Head generates the final object predictions, including class labels and bounding box coordinates.

Among the different versions of YOLOv5 (including YOLOv5s, YOLOv5m, and YOLOv5l) [9], we compared their effectiveness in terms of precision, inference speed, and resource consumption. After thorough evaluation, we chose YOLOv5s, the smallest variant of YOLOv5 [9]. Although it offers a slight trade-off in accuracy compared to larger versions, YOLOv5s significantly improves inference speed and reduces computational requirements. This makes it the optimal choice for real-time applications, where speed and efficiency are critical.

YOLOv5s has found widespread adoption in various domains, including livestock monitoring. In agricultural applications, particularly for real-time monitoring of animals, YOLOv5s proves to be highly effective in processing high-resolution images quickly, making it ideal for large-scale farms where numerous animals need to be monitored simultaneously. Its capability to identify and track livestock in real-time has significantly enhanced monitoring systems, improving productivity, animal welfare, and disease detection in smart farming systems.

B. *PLM-YOLOv5*

The framework of PLM-YOLOv5 is illustrated in Figure 2, where we present an enhanced version of the YOLOv5 [9] architecture designed specifically to handle the task of detecting pigs in livestock monitoring. The core of the framework revolves around modifying the original YOLOv5 [9] model to cater to the unique challenges posed by our dataset, which consists of more than 1,400 annotated photos.

1) Modified YOLOv5 Architecture: YOLOv5 [9], a cutting-edge model for real-time object detection, is modified in PLM-YOLOv5 to improve its performance on our specific dataset. The primary change is in the prediction head, which has been optimized to detect small pigs more effectively. In many livestock scenarios, pigs are often small or occluded, making them difficult to detect using standard object detection models. Therefore, the prediction head in PLM-YOLOv5 has been adapted to ensure higher sensitivity to smaller objects and better localization accuracy for pigs.

2) Channel Attention Mechanism (CBAM): To enhance the model's performance, we incorporated the Convolutional Block Attention Module (CBAM) into the YOLOv5 [9] backbone. This lightweight attention mechanism improves the model's feature representation by emphasizing the most important regions of the input image. CBAM applies two attention mechanisms: spatial attention and channel attention. The spatial attention mechanism highlights "where" the important features are in the image, while the channel attention mechanism emphasizes "what" features are most relevant. By incorporating CBAM, PLM-YOLOv5 is better able to handle the challenges posed by varying lighting conditions, occlusions, and the presence of multiple pigs in complex farm environments.

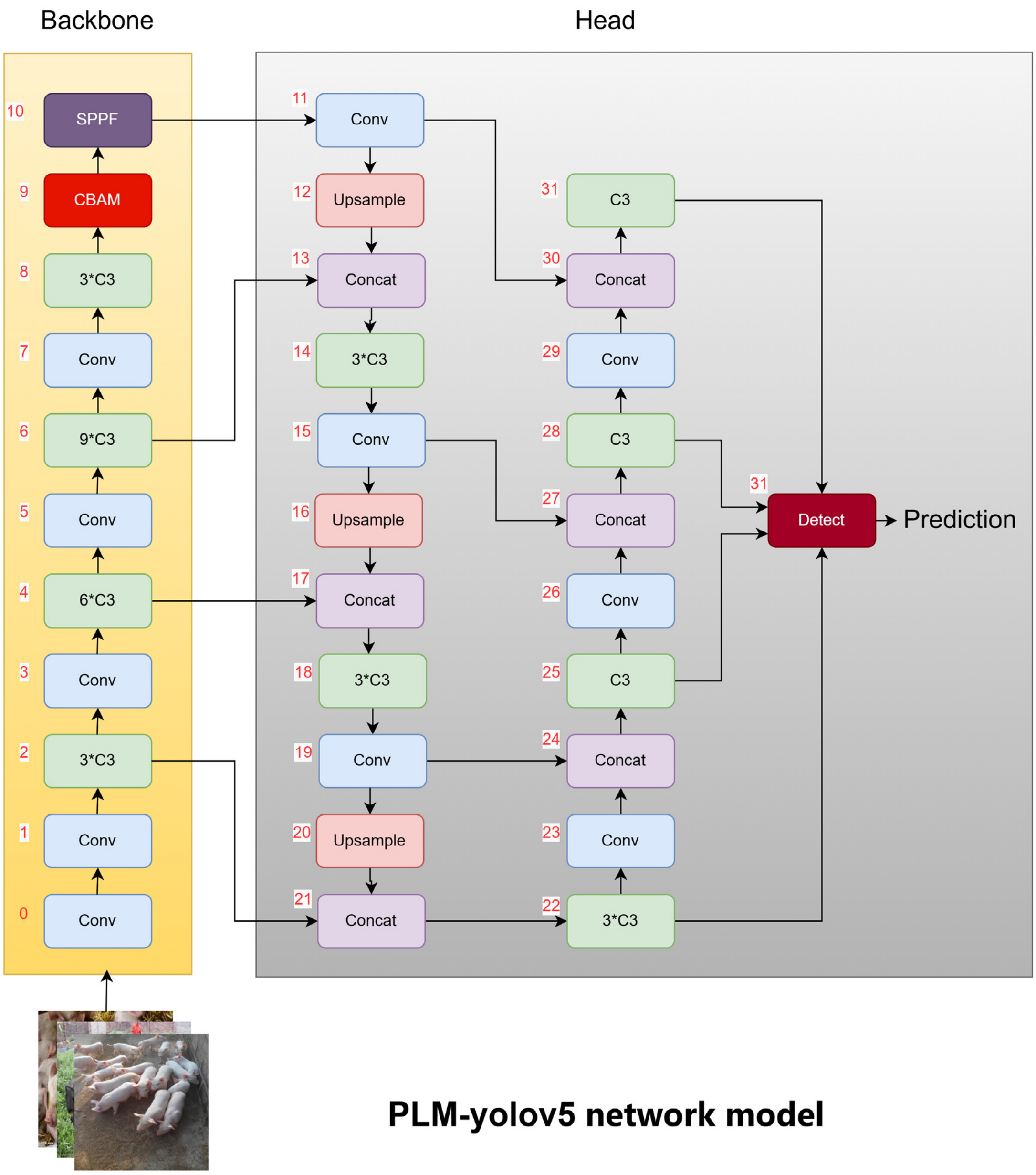

Figure 2. PLM-YOLOv5 Model Architecture. This figure illustrates the architecture of PLM-YOLOv5, showcasing the integration of the Small Object Detection Head and the CBAM (Convolutional Block Attention Module). These modifications are designed to enhance small object detection and improve feature learning for more accurate and robust object detection.

3) Dataset Construction and Annotation Methodology: The success of any object detection model depends heavily on the quality of the dataset used for training. For PLM-YOLOv5, the dataset consists of over 1,400 annotated images that capture pigs in various settings and conditions. These images were carefully curated from diverse environments, ensuring that the model would be robust enough to handle different types of scenes encountered in real-world farm conditions. The annotation process was carried out using LabelImg, a popular open-source graphical image annotation tool. LabelImg allows users to draw bounding boxes around objects of interest and assign them labels, making it suitable for annotating images for object detection tasks. Each pig in the images was labeled with the class "pig" and bounding boxes were carefully drawn to enclose the pig in every frame. Special attention was given to ensure that small pigs, which are more difficult to detect, were accurately annotated. To address this challenge, smaller bounding boxes were used for these objects, and extra effort was made to ensure the consistency and accuracy of annotations across the entire dataset. An sample from the dataset with annotations is illustrated in Figure 3, where the bounding boxes clearly indicate the locations of the pigs.

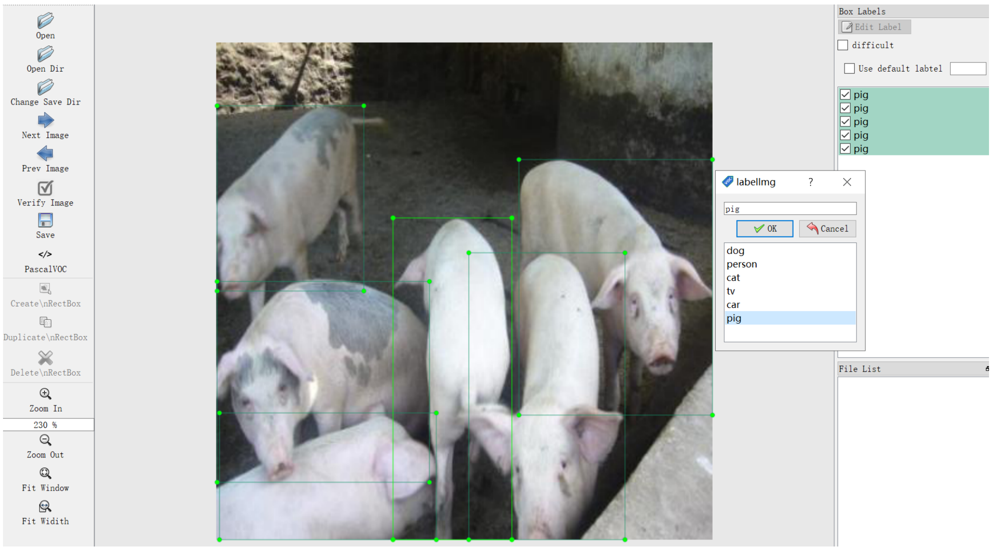

Figure 3. Labeling Demo with LabelImg. This figure shows a demonstration of using LabelImg for annotating images in a dataset. The tool allows for easy and efficient labeling of objects by drawing bounding boxes around targets, which are essential for training object detection models. The image illustrates the process of labeling, where the user selects object categories and adjusts the bounding boxes to fit the objects in the image.

IV. EXPERIMENTS

A. *Implementation Details*

The dataset used for training, validation, and testing consists of 1137 training samples, 315 validation samples, and 10 test videos. The training set was used to optimize the model, while the validation set was used to adjust hyperparameters and mitigate overfitting. The 10 test videos were employed to assess the model's performance in a real-world setting, focusing on its ability to detect objects across multiple frames. The input images were resized to 416x416 pixels to maintain consistency and ensure optimal performance. This image size is commonly used in YOLOv5 [9] for balancing accuracy and processing speed. Data augmentation techniques, including random flips, scaling, and cropping, were utilized to enhance the diversity of training samples and strengthen the model's robustness.

The model was trained on an ecs.gn7i-c8g1.2xlarge instance, which includes an NVIDIA A10 GPU, 8 CPU cores, and 30 GiB of RAM. The use of the A10 GPU provided significant acceleration for the model training, allowing for faster processing and improved performance compared to using a CPU-only setup. The powerful GPU was particularly beneficial when training the model on large datasets and when using complex deep learning architectures like YOLOv5 [9]. The model was trained for 300 epochs with a batch size of 16. A learning rate scheduler was employed to progressively reduce the learning rate during training, while the Adam optimizer adapted the learning rate for each parameter, ensuring faster convergence and enhanced performance.

The training was conducted using PyTorch 2.5.1+cu124, which enabled efficient GPU utilization and supported the necessary features for model optimization and training. The framework also facilitated easy integration of custom layers and modules, such as the CBAM attention mechanism and small-object detection heads, which were tested in the ablation experiments.

B. *Comparisons*

Based on the experimental results of PLM-YOLOv5, the model achieved a mAP@0.5 of 0.76 and a mAP@0.5:0.95 of 0.9. These results demonstrate strong performance, particularly in scenarios where the overlap requirement is more lenient, as indicated by the higher mAP@0.5:0.95. The relatively higher mAP@0.5 suggests that PLM-YOLOv5 is capable of providing better object localization compared to other models, making it more reliable for precise detection tasks.

When comparing PLM-YOLOv5 with other state-of-the-art models like YOLOv5 [9], YOLOv4 [21], and Faster R-CNN [16], PLM-YOLOv5 shows competitive performance. It outperforms YOLOv5 [9] (0.67) and YOLOv4 [21] (0.7) in both mAP@0.5 and mAP@0.5:0.95. The model achieves a significant improvement in detection accuracy, with a mAP@0.5 of 0.76 and a mAP@0.5:0.95 of 0.9, making it the best-performing model in this comparison. Despite its impressive performance, Faster R-CNN [16] achieves a high mAP@0.5 (0.75) but falls behind in mAP@0.5:0.95 (0.78), highlighting its limitations in handling varying levels of overlap between detected objects.

The improvements in PLM-YOLOv5 are attributed to the enhanced position-localization module, which aids in detecting small or occluded objects more effectively. However, further optimization is still needed to enhance its performance on stricter IoU thresholds. Table I shows the performance comparison of different models.

TABLE I. PERFORMANCE COMPARISON OF DIFFERENT MODELS

Model	mAP@0.5 (%)	mAP@0.5:0.95 (%)
PLM-YOLOv5	76	90
YOLOv5	67	80
YOLOv4	70	78
Faster R-CNN	75	78

C. Experiments and Analysis

The inference performance of the model was evaluated on a video from the PigData test set, specifically the Y-18.mp4 video. The model processed a total of 2293 frames, each resized to a resolution of 384x640 pixels. On average, the model detected 6 pigs per frame, demonstrating its effectiveness in identifying the target objects.

The inference speed was measured at approximately 5.9ms per frame, indicating a fast and efficient detection process suitable for real-time applications. The model's pre-processing time was 0.2ms, which includes image resizing and normalization, while the Non-Maximum Suppression (NMS) step took 0.8ms. These processing times demonstrate that the model is highly optimized for rapid inference, with the majority of the time spent on the actual object detection process. Figure 4 shows a sample frame from the Y-18.mp4 video with the pigs detected and highlighted. Given these performance metrics, the model is well-suited for real-time video analysis tasks, where both speed and accuracy are critical.

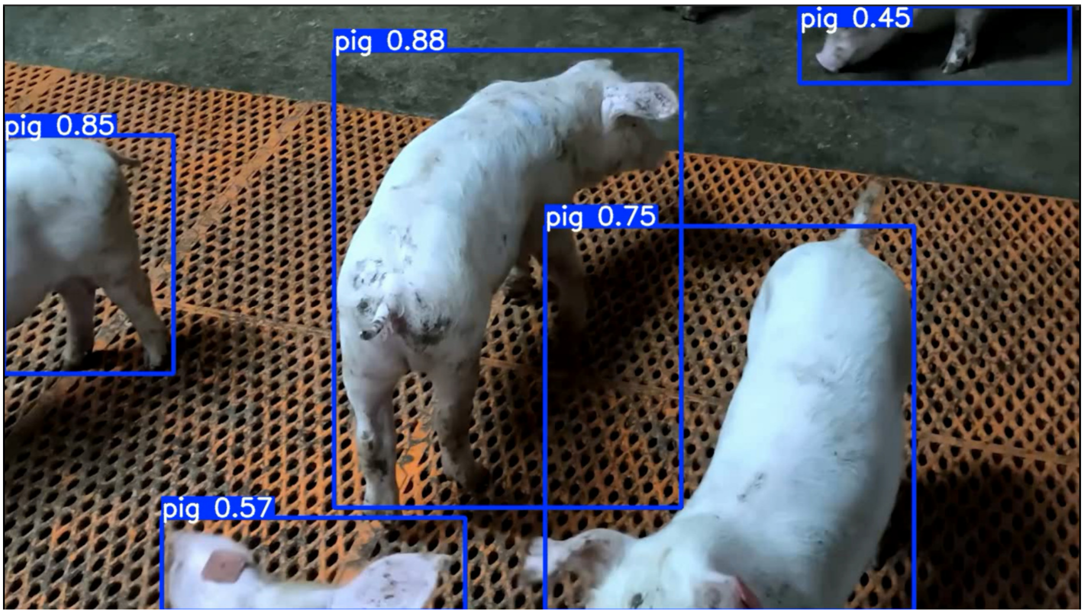

Figure 4. Pig Detection in Video Frame. This figure shows the detection results of pigs in a sample frame from the Y-18.mp4 video, where the model successfully identifies and highlights the pigs within the scene.

D. Ablation Studies

The performance of different configurations of PLM-YOLOv5 models was evaluated through ablation experiments, focusing on the impact of specific modifications on the model's accuracy.

YOLOv5s achieved a mAP of 0.676, serving as the baseline model. Introducing a Small Object Detection Head in YOLOv5s+Small Object Detection Head slightly improved performance to 0.689, indicating that the model benefits from additional capabilities for detecting small objects. When a CBAM (Convolutional Block Attention Module) was integrated into YOLOv5+CBAM, the model showed a further improvement with a mAP of 0.712, demonstrating the effectiveness of attention mechanisms in enhancing feature representation. Finally, PLM-YOLOv5, which combines the enhancements from both small object detection and CBAM, achieved the highest mAP of 0.764, showcasing the significant improvements in both accuracy and robustness.

The ablation studies confirm that each added component, whether targeting small object detection or improving feature learning with CBAM, contributes positively to the overall model performance. As shown in Table II, the performance of each model variant clearly demonstrates the positive impact of the enhancements introduced in each ablation step.

TABLE II. ABLATION STUDY RESULTS

Method	**Model (M)**	**Param (M)**	**FLOPs (G)**	**FPS (frames/ second)**	**mAP (%)**
YOLOv5s	13.7	7.01	15.8	3.99	67.6
YOLOv5s+Small Object Detection Head	14.7	7.02	16.9	3.95	68.9
YOLOv5+CBAM	13.8	7.11	17.8	3.9	71.2
PLM-YOLOv5	14.7	7.19	18.5	3.71	76.4

V. CONCLUSIONS

In this study, PLM-YOLOv5 demonstrated significant improvements in object detection accuracy, particularly for small objects, when compared to other models such as YOLOv5 [9], YOLOv4 [21], and Faster R-CNN [16]. The introduction of the Small Object Detection Head and the CBAM (Convolutional Block Attention Module) contributed to the model's enhanced performance, achieving a mAP of 0.764. This result highlights the effectiveness of these modifications in improving both detection precision and robustness, particularly in real-time applications.

However, despite the promising results, there are still areas for improvement. While PLM-YOLOv5 outperforms other models in detection accuracy, further optimization is needed to refine localization accuracy, especially under stricter IoU thresholds. Additionally, while PLM-YOLOv5 maintains competitive FPS, there is potential for further acceleration, especially for deployment on resource-constrained devices. Future work will aim to overcome these limitations and enhance the model's efficiency for real-time detection applications.

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